

Muzzammil Ahmed

+91 8109418626 | muzzammil.cse3@gmail.com

[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Bhopal, Madhya Pradesh, India

PROFESSIONAL SUMMARY

M.Tech Computer Science Engineering student with hands-on experience in Machine Learning, Deep Learning, Python, and Data Analytics. Experienced in developing ML/DL models, CNN-based computer vision applications, RAG-based solutions, and Streamlit applications. Skilled in data preprocessing, feature engineering, model optimization, and performance evaluation. Currently conducting research on hybrid machine learning models for heart disease prediction.

EDUCATION

All Saints' College of Technology <i>M.Tech in Computer Science Engineering CGPA: 7.63</i>	Aug 2024 – Expected Dec 2026 Bhopal, India
All Saints' College of Technology <i>B.Tech in Computer Science Engineering CGPA: 8.32</i>	Jun 2020 – Jul 2024 Bhopal, India
Royal Higher Secondary School <i>Higher Secondary (PCM) 63%</i>	2018 – 2020 Bhopal, India
Kendriya Vidyalaya No.3 <i>High School 62%</i>	2008 – 2018 Bhopal, India

Skills

Programming: Python, SQL, JavaScript

Data Analytics: Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization, Statistical Analysis, Feature Engineering

Machine Learning: Regression, Classification, Decision Trees, KNN, Ensemble Learning, Feature Selection, Hyperparameter Tuning, Cross-Validation, SMOTE

Deep Learning & AI: ANN, CNN, NLP, Transformers, RAG, LangChain

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, OpenCV, Streamlit, Matplotlib, Seaborn

Frontend Development: HTML, CSS, React.js

Business Intelligence: Power BI, Excel, Pivot Tables, Charts, Slicers, KPIs, Dashboard Development

Databases: MySQL

Tools: Git, GitHub, Jupyter Notebook, VS Code, Figma, Netlify

EXPERIENCE

Muslim Education and Career Promotion Society (MECAPS) <i>AI/ML Training Intern</i>	Jul 2025 – Jun 2026 Bhopal, India
- Worked on Machine Learning and Deep Learning concepts through practical training and project-based learning. - Developed ML models using Python, Pandas, NumPy, Scikit-learn, TensorFlow, and Keras. - Performed data preprocessing, exploratory data analysis, feature selection, model training, and model evaluation. - Worked with ANN, CNN, NLP, and hybrid machine learning approaches on real-world datasets.	

RESEARCH

Concatenated Hybrid Ensemble Model for Heart Disease Prediction *Under Review*

- Designed 5 hybrid architectures and 14 concatenated ensemble combinations using Logistic Regression and Decision Tree as base learners, inspired by concatenated hybrid ensemble voting approaches in literature.
- Best single-split model achieved 91.80% accuracy, 92.54% F1-score, 91.18% precision, and 93.94% recall on the Cleveland heart disease dataset.
- Implemented a leakage-free 10-fold cross-validation protocol with McNemar's test and repeated 5x2 cross-validation for statistical rigor; best cross-validated model (Extra-Trees-selected hybrid) achieved 85.41% accuracy, 87.22% F1-score, and 90.66% AUC.

PROJECTS

PDF Chatbot using RAG and LangChain [🔗 GitHub](#)

- Developed a Retrieval-Augmented Generation (RAG) chatbot for question answering over uploaded PDF documents.
- Implemented document loading, text chunking, embeddings, vector storage, and semantic retrieval for relevant context.
- Integrated LangChain with an LLM to generate context-aware responses based on retrieved document content.
- Implemented a Streamlit interface for interactive document-based question answering.

Delivery Time Prediction System [🔗 GitHub](#)

- Developed an end-to-end delivery time prediction system using the Porter logistics dataset.
- Performed exploratory data analysis, feature engineering, data preprocessing, and statistical analysis to identify key factors affecting delivery time.
- Built and optimized an ANN regression model using TensorFlow with hyperparameter tuning and model evaluation.
- Developed a Streamlit web application for real-time delivery time prediction using the trained model.

Face Mask Detection using CNN [🔗 GitHub](#)

- Developed a Convolutional Neural Network (CNN) model for classifying images as with-mask or without-mask.
- Performed image preprocessing and data preparation using OpenCV and NumPy for model training.
- Trained and evaluated the CNN model using TensorFlow and Keras for image classification.
- Built a Streamlit application for real-time face mask detection using the trained CNN model.

CERTIFICATIONS

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| • Python - Introduction to Data Science and Machine Learning A-Z | View Certificate |
| • Python for Data Analytics - Beginner to Advanced | View Certificate |
| • Learn MySQL - For Beginners | View Certificate |
| • Machine Learning Basics - Regression Analysis | View Certificate |

ADDITIONAL INFORMATION

Languages: English, Hindi

Areas of Interest: Artificial Intelligence, Machine Learning, Data Analytics, Natural Language Processing, Computer Vision

Strengths: Problem Solving, Analytical Thinking, Fast Learner, Research-Oriented Mindset, Adaptability